Evaluating Quantized Vision-Language Models for Spacecraft-Relevant Inference on Constrained Hardware

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***Abstract*—Vision-Language models (VLMs) have emerged as a promising tool for enabling spacecraft to interpret visual data and make autonomous decisions. However, due to spacecraft hardware constraints, including limited compute capacity, memory, and network connectivity, deploying high-quality models onboard remains a significant challenge. We present an evaluation of VLM performance in environments simulating spacecraft constraints, benchmarking the capabilities of compact models across various use cases. We further explore the challenges encountered in the process and how they can be overcome to enable VLM inference on future spacecraft.**

***Index Terms*—vision-language model (VLM), machine learning, high-performance onboard computing, software development**

I. Introduction

This study addresses the challenge of enabling vision-language models (VLMs) to operate on resource-constrained spacecraft hardware. [7] Onboard VLM inference may allow spacecraft to interpret visual data locally, reducing the time, bandwidth, and human analysis resources required for selected mission tasks. Although VLMs offer a promising capability for spacecraft visual interpretation, their model size and computational demands currently exceed the capabilities of many space-ready computers.

Exploring deep space requires spacecraft to operate with greater autonomy due to long communication delays and limited human oversight. Previous spacecraft autonomy efforts, such as the EO-1 Autonomous Sciencecraft, demonstrated that onboard analysis, planning, and execution can enable a spacecraft to detect and respond to science events while operating under limited computing resources [2]. Current spacecraft computers are significantly limited compared to Earth-based systems in terms of memory, processing power, and energy availability, making it difficult to deploy advanced artificial intelligence systems. These constraints motivate the evaluation of compact models and efficient inference tools that can operate within realistic onboard computing limits.

This work is supported by NASA Jet Propulsion Laboratory (JPL) at the California Institute of Technology, as well as by the Department of Computer Science at the University of Colorado Boulder.

Previous studies have shown that vision-language models are a promising class of models for image reasoning and natural-language visual interpretation in space domains [1]. VLMs can generate descriptions, answer questions about images, and produce structured outputs from visual input, suggesting potential utility for spacecraft visual decision support. Unlike conventional image classification models, VLMs can combine visual inputs with natural-language prompts to produce flexible task-specific responses. This capability may be useful for spacecraft operations such as terrain description, anomaly assessment, science target prioritization, or structured status reporting. However, this flexibility also introduces new evaluation challenges, since model outputs may vary across runs and may require assessment of both factual visual accuracy and response consistency. Additionally, support for image processing makes these models computationally expensive, which presents new challenges given the limitations of space-qualified hardware.

Therefore, there is a critical need to identify, adapt, and evaluate lightweight VLM architectures that can function reliably within the harsh constraints of spacecraft environments. Candidate models must be able to run efficiently on limited onboard computing hardware, provide accurate and useful interpretations of planetary or spacecraft imagery, and support reliable local inference under strict compute and memory limitations. For mission operations, these outputs must also be task-relevant, interpretable, and consistent enough to support human review.

This work evaluates selected compact VLMs on a single-board computer as a proxy to simulate a hardware-constrained environment. Candidate models are tested across a range of space-relevant scenarios to evaluate model runtime performance and output quality across various cases. This study provides an assessment of whether small VLMs can provide useful visual interpretation under strict compute constraints, along with model-selection guidance by examining tradeoffs between inference speed, and output quality. This paper makes the following contributions:

1) Evaluates compact VLM inference on constrained

single-board computing hardware for space-relevant visual interpretation tasks.
2) Captures runtime performance using token throughput, and wall time.
3) Provides model selection guidance under runtime, memory, output-quality, and U.S. government contract constraints.

II. Background

A. Quantization and Local Inference

Large language and vision-language models are typically trained and deployed using high-precision numerical formats and high-performance processors. These assumptions do not hold for spacecraft-relevant computing environments where memory, power, and processor availability are limited. Local inference in these environments requires models and runtimes that can operate within strict resource limits.

Quantization offers a potential solution by reducing the numerical precision used to represent model parameters, lowering memory requirements [4]. Quantized model formats reduce the precision used to represent model parameters, allowing larger models to fit within limited memory constraints and reducing memory requirements during local inference. This optimization allows for capable models to be run on hardware that would otherwise be insufficient, making it possible to run advanced VLM inference in the resource-constrained environments relevant to this study.

This study did not train new models, fine-tune existing weights, or create new quantized variants. Instead, it evaluates existing compact VLM variants and available quantized model formats as candidates for constrained local inference. The models are executed locally using llama.cpp libraries, which supports inference for VLMs on commodity hardware. Local execution is necessary as onboard inference cannot assume connectivity in spacecraft scenarios.

B. Evaluating VLM Outputs

Evaluating VLM outputs differs from evaluating traditional classifiers because responses are open-ended and task-dependent. A response might be partially correct, visually accurate but scientifically incomplete, or well organized but containing incorrect interpretations. Therefore, responses must be comprehensively evaluation using various factors to capture the overall quality of the output.

Recent work has explored the use of large language models as evaluators for open-ended model responses [5]. Such methods can provide scalable comparative scoring when expert-labeled ground truth is not available. However, they do not replace assessment from domain experts, and their results should be interpreted as relative quality estimates rather than objective scientific correctness. For spacecraft and planetary science applications, expert validation remains necessary for final model qualification.

III. Methods

A. Tools and Technologies

1) Experimental Platform: All experiments were run on a Raspberry Pi 5 with 16 GB of memory. This device was selected as a low-power single board computer representing that of onboard inference. Although the Raspberry Pi is not equivalent to space-qualified hardware, it provides a practical test interface for evaluating CPU only VLM inference.

2) Inference Software: A C++ command-line interface was developed to manage local VLM inference through llama.cpp. The interface supports text-only and image-conditioned prompts, model configuration through command-line flags, and automated execution across multiple model variants. For image-based inference, the system uses a vision projector model in conjunction with the core language model. The tool was designed to simplify repeated benchmarking and to provide a consistent interface to evaluate model behavior across prompts, images, and quantization formats.

B. Models

The evaluation included consisted of quantized variants from the Gemma 3 and Ministral 3 model families. Model sizes ranged from 3B to 27B variants. Multiple quantization strategies were tested to assess the relationship between compression, runtime performance, and response quality.

Table I summarizes the evaluated models.

TABLE I
Tested Models and Quantization Strategies

Family	Parameter Count	Quantization
Gemma 3	4B	4-bit QAT
Gemma 3	4B	4-bit K_M
Gemma 3	4B	8-bit
Gemma 3	12B	4-bit QAT
Gemma 3	12B	4-bit K_M
Gemma 3	12B	8-bit
Gemma 3	27B	1-bit IT, UD
Ministral 3	3B	4-bit QAT
Ministral 3	3B	4-bit K_M
Ministral 3	8B	4-bit QAT
Ministral 3	8B	4-bit K_M
Ministral 3	14B	4-bit QAT
Ministral 3	14B	4-bit K_M

Note: QAT = Quantization Aware Training; K_M = K-Quants (Medium) Quantization; IT = Instruction-Tuned; UD = Unsloth Dynamic Quantization.

C. Testing

1) Test Prompts: The models were evaluated using image-prompt pairs based on space-relevant visual interpretation scenarios. The test set included planetary images, terrain imagery from Mars, and rover wheel imagery for damage assessment. Prompts were designed to cover both open-ended interpretation and structured-output tasks. Open-ended prompts assessed scene description and scientific reasoning, while structured prompts evaluated whether models could produce constrained outputs such as grid-based damage ratings. Structured prompts

were included because spacecraft software often benefits from outputs that can be directly parsed and used elsewhere.

This task set was intended to evaluate model behavior across a range of plausible onboard visual analysis use cases rather than to provide a definitive benchmark. This measures not only whether a model could describe the image, but also whether it could produce outputs that are useful, consistent, and operationally relevant. The prompts were held constant across model variants to support comparative analysis.

2) Performance Metrics: During testing, the following metrics were captured to evaluate model performance:

- **Tokens per second:** Number of tokens generated per second.
- **Prompt tokens:** Number of tokens in the input prompt sent to the model.
- **Gen tokens:** Number of tokens generated by the model in the response.
- **Total tokens:** Sum of prompt tokens and generated tokens.
- **Prompt ms:** Time taken to process the input prompt.
- **Gen ms:** Time taken to generate the output tokens.
- **Time to first token (ttft) ms:** Delay before the first output token is produced after sending the prompt.
- **Wall sec:** Total elapsed real-world time in seconds from prompt to response.

D. Evaluation Protocol

Model output quality was evaluated using a comparative LLM-as-a-judge approach [6]. For each prompt, Gemini 3.1 Pro was given the prompts along with additional metadata to create a 'gold standard' output that was used as the reference source when evaluating other models. This process provided insight into the expectations for the most capable models available compared to those explored in this study. The prompts and evaluation techniques were based on previous research on the use of LLMs to judge the results of the model [5]. Output quality was evaluated using a structured rubric with the following five criteria:

- **Visual Accuracy** (V)**:** Does the model correctly describe what is visually present in the image?
- **Scientific Reasoning** (S)**:** Does the model demonstrate understanding of the science behind what it sees?
- **Investigation Quality** (I)**:** Are the proposed investigations or follow-up questions specific, feasible, and scientifically motivated?
- **Completeness** (C)**:** How much of the relevant content does the response cover?
- **Coherence and Organization** (O)**:** Is the response well-structured, readable, and logically organized? Does information flow naturally?

The rubric and model outputs were provided to another frontier model, Claude Opus 4.6, which graded each criterion a scale from 1 to 5 based on guidelines outlined in the rubric. A weighted composite score was computed as

$$\text{score} = 0.3V + 0.3S + 0.20I + 0.15C + 0.05O \qquad (1)$$

where V is visual accuracy, S is scientific reasoning, I is investigation quality, C is completeness, and O is coherence. The weight of each component was decided on the relative importance and contribution to overall output quality, prioritizing correct visual interpretation and scientific reasoning as the most important qualities for a complete output.

Because expert-labeled planetary-science ground truth was not available for each of our prompt-image pairs, model outputs were compared against reference responses generated by a frontier model. A separate frontier model was then used to assign rubric scores. This approach provides a scalable comparative evaluation across topics, but does not establish objective scientific correctness. The resulting scores should therefore be interpreted as relative indicators of output quality rather than as substitutes for expert validation.

IV. Results

A. Runtime Performance

Table II summarizes representative runtime results for the evaluated models. Fig. 1 presents the average total time from prompt to response for all VLMs evaluated.

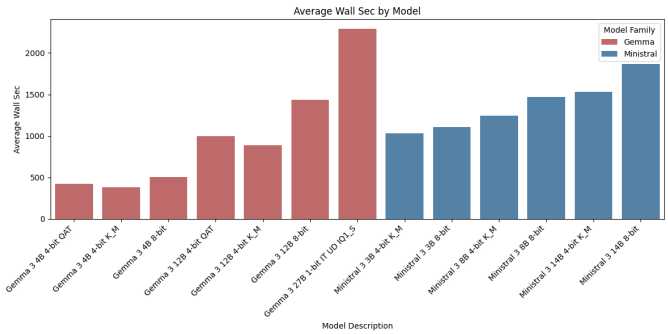

Fig. 1. Comparison of wall time to response across models

B. Model Output Quality

Fig. 2 presents the average composite scores for all VLMs evaluated.

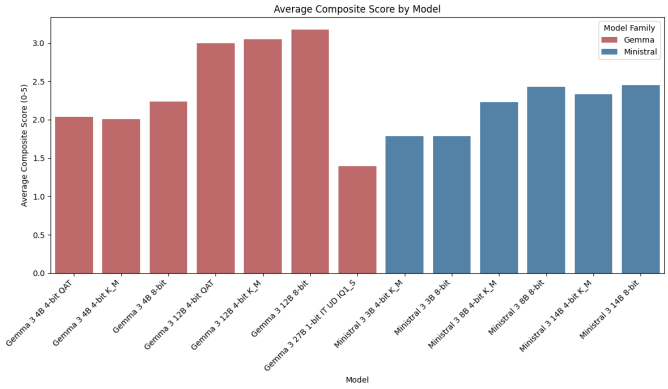

Fig. 2. Comparison of composite scores across models.

TABLE II
PERFORMANCE COMPARISON OF MODELS

Model	tok/sec	prompt tok	gen tok	total tok	prompt ms	gen ms	ttft ms	wall sec
Gemma 3 4B 4-bit QAT	3.51	330	756	1086	202747	216352	1	419.70
Gemma 3 4B 4-bit K_M	4.38	328	726	1054	217362	167137	1	385.12
Gemma 3 4B 8-bit	2.69	328	778	1106	213952	289275	1	503.86
Gemma 3 12B 4-bit QAT	1.29	330	947	1277	266224	730224	1	997.47
Gemma 3 12B 4-bit K_M	1.46	328	819	1147	323447	563566	1	887.74
Gemma 3 12B 8-bit	0.80	328	880	1208	310115	1125102	1	1436.16
Gemma 3 27B 1-bit IT UD IQ1_S	0.60	328	568	896	1383724	905366	1	2289.62
Ministral 3 3B 4-bit K_M	3.59	1096	284	1380	950247	82741	1	1033.20
Ministral 3 3B 8-bit	2.47	1096	342	1438	966286	142721	1	1109.24
Ministral 3 8B 4-bit K_M	1.85	1096	333	1429	1053535	187347	1	1241.16
Ministral 3 8B 8-bit	1.09	1096	409	1505	1096948	375012	1	1472.32
Ministral 3 14B 4-bit K_M	1.19	1096	414	1510	1179908	347496	1	1527.77
Ministral 3 14B 8-bit	0.71	1096	442	1538	1255562	610740	2	1866.75

V. DISCUSSION

A. Model Evaluation

As expected, the larger parameter models took more total time (wall sec) and generated tokens at a slower rate. The fastest configuration was Gemma 3 4B 4-bit K M, which achieved 4.38 tokens/sec with a wall time of 385.12 seconds. Upon examination of the performance metrics and composite scores, most of the models performed as expected relative to the other models tested in this study. There were a few exceptions, with the most surprising result being the average composite score for the 1-bit Gemma 3 27B model. While this model generated output the slowest at 0.6 tokens/sec, it had the lowest average composite score of 1.4. Further investigation showed that there was no single composite score skewing the average. The model scored a max of 2.08 for PCat7 and scored below 2 for all other test prompts. This was the largest model tested in this study, and its poor performance shows the impact of extreme compression on overall model performance. Extreme quantization significantly reduces the numerical precision used to represent model weights, which can lead to substantial information loss during inference. As precision decreases to very low bit-widths such as 1-bit, the model becomes less capable of preserving the complex relationships learned during training, resulting in reduced accuracy, weaker reasoning ability, and lower quality outputs.

The results also show a big tradeoff in output speed as models grow larger. Smaller 4-bit models, like the Gemma 3 4B and Ministral 3B, produced tokens at a much faster rate. For example, the Gemma 3 4B 4-bit K_M model generated tokens at a rate of 4.38 tokens/sec on average, which is approximately 3 times faster than its 12B variant at 1.46 tokens/sec, making these smaller models much more convenient for tasks that require quick feedback. Additionally, the Mistral models used significantly more prompt tokens than the Gemma models even when the models had fewer parameters. Overall, both model families showed potential value in the scenarios studied, where Gemma models generally received a higher average composite score than Mistral models of comparable size.

B. Limitations

Despite these encouraging results, several limitations should be acknowledged when interpreting our findings:

- **Evaluation methodology:** Model outputs were evaluated against responses generated by a frontier LLM rather than domain expert interpretations. While this provides a baseline suitable for our performance benchmarking needs, it does not capture the expertise of a trained geologist or mission scientist.
- **Hardware constraints:** All inference in this work was performed on the Raspberry Pi 5 CPU without GPU acceleration. Although this still produced noteworthy results, the lack of GPU acceleration did impose a ceiling on performance expectations and it is likely that greater performance can be achieved using the GPU. Furthermore, GPU acceleration may allow for larger and more capable models to be explored.
- **Unexpected performance of larger models:** Some larger quantized models showed unexpected weak performance, such as the 1-bit Gemma 3 27B model, indicating that aggressive compression can significantly reduce a model's usefulness despite the large parameter count. Due to hardware limitations, this study could not evaluate additional quantizations for larger models, which may provide high quality results on more capable systems.
- **Lack of domain-specific training:** The models evaluated were general-purpose VLMs with no fine-tuning on space-specific imagery or terminology. Performance on highly domain-specific inputs may differ substantially on models with domain-specific knowledge.

C. Recommended Model Selection

Through considering both the performance metrics and output quality, the models can be evaluated to balance output quality with performance requirements in various scenarios. The following is an evaluation of the models and their scores with different constraints that may be encountered.

1) Model Recommendations by Response Time: Table III summarizes recommended models based on response time constraints.

TABLE III
SMALL VLM SELECTION BASED ON RESPONSE TIME CONSTRAINTS

Time Requirement	Model	Wall Time	Score
<8 min	Gemma 3 4B 4-bit QAT	<7 min	2.04
<8 min	Gemma 3 4B 4-bit K_M	<6.5 min	2.01
<10 min	Gemma 3 4B 8-bit	<8.5 min	2.24
<15 min	Gemma 3 12B 4-bit K_M	<15 min	3.05
<25 min	Gemma 3 12B 8-bit	<24 min	3.18

Note: Due to the hardware constraints of the Raspberry Pi 5 and the lack of a dedicated GPU on the single-board computer, this work does not explore models running with GPU acceleration. All processing was performed using only the CPU. The times found during testing were impacted by this limitation, and using a dedicated GPU will significantly decrease time while allowing larger models to be run.

2) Model Recommendations by Use Case: Table IV lists top-performing models for each use case explored in the test set based on composite scores.

TABLE IV
SMALL VLM SELECTION BY USE CASE BASED ON COMPOSITE SCORE

Use Case	Model	Score
Planetary Analysis	Gemma 3 12B 4-bit K_M	3.3167
Planetary Analysis	Gemma 3 12B 8-bit	3.2750
Damage Assessment	Gemma 3 12B 8-bit	2.9000
Damage Assessment	Gemma 3 12B 4-bit QAT	2.6250
Terrain Analysis	Gemma 3 12B 8-bit	3.1667
Terrain Analysis	Gemma 3 12B 4-bit K_M	2.9833

3) Model Recommendations by Scoring Rubric Metrics: Table V summarizes the top two models for each rubric metric.

TABLE V
SMALL VLM SELECTION BY SCORING RUBRIC METRICS

Metric	Model	Score
Visual Accuracy	Gemma 3 12B 8-bit	3.2857
Visual Accuracy	Gemma 3 12B 4-bit K_M	3.2143
Scientific Reasoning	Gemma 3 12B 8-bit	3.2143
Scientific Reasoning	Gemma 3 12B 4-bit K_M	3.0000
Investigation Quality	Gemma 3 12B 8-bit	2.8571
Investigation Quality	Gemma 3 12B 4-bit QAT	2.7143
Completeness	Gemma 3 12B 8-bit	3.0000
Completeness	Gemma 3 12B 4-bit K_M	2.9286
Coherence	Gemma 3 12B 4-bit K_M	4.0714
Coherence	Gemma 3 12B 8-bit	4.0000

Note: This evaluation is based on a custom scoring rubric made specifically for this study, designed to be used on untrained models. This evaluation does not necessarily reflect the performance of the same models in each of the above categories after being trained using domain-specific material.

VI. CONCLUSION

A. Summary

This study demonstrates that VLMs can be deployed on limited hardware while still producing useful outputs for space-related geological analysis, weather pattern analysis, and spacecraft damage assessments. Using a Raspberry Pi 5 as the sole inference platform, our results show that even relatively large models such as the 12B parameter Gemma 3 variants can produce coherent, relevant outputs within practical time limits. This finding is significant in the context of space exploration, where communication delays with Earth make onboard autonomous analysis a crucial capability.

B. Future Work

Several directions could meaningfully extend this work. With stronger onboard hardware and by enabling GPU acceleration, one can expect even faster inference speeds, and the use of larger, more capable models should become significantly more practical. Extending the benchmarking framework to include a wider range of RISC-V based processors, such as the PolarFire and BeagleBoard platforms, would provide a more comprehensive evaluation of VLM performance across embedded systems [3].

Given that model training was out of the scope of this study, such fine-tuning or parameter-efficient adaptation of existing VLMs on relevant space geology and mission imagery datasets would likely yield substantial gains in domain-specific accuracy. Additionally, replacing the frontier LLM evaluator with structured expert annotation would strengthen the validity of future performance benchmarks. Future studies may also explore other high-performing model families, many of which were not tested because of licensing or policy concerns, preventing their use for NASA projects.

C. Closing Remarks

Overall, this work supports the feasibility of onboard VLM inference as a practical tool for autonomous spacecraft operations. Deploying capable, domain-adapted vision-language models directly onto deep space spacecraft represents a promising avenue for reducing dependence on Earth communication links and enabling more responsive, intelligent mission operations.

ACKNOWLEDGMENT

The authors thank NASA Jet Propulsion Laboratory and the University of Colorado Boulder for supporting this work.

REFERENCES

[1] A. Carrasco *et al.*, "Visual Language Models as Operator Agents in the Space Domain," 2025. [Online]. Available: https://arxiv.org/html/2501.07802v1

[2] S. Chien *et al.*, "Lessons Learned from the Autonomous Sciencecraft on Earth Observing One," 2006. [Online]. Available: https://www.stsci.edu/largefiles/iwpss/2006620252chien-s-iwpss2006.pdf

[3] P. Ghiglino, M. Harshe, H. D. Doran, and C. R. Tordoya Taquichiri, "High-Performance AI Inference for Agile Deployment on Space-Qualified Processors: A Performance Benchmarking Study," in *Computer Safety, Reliability, and Security. SAFECOMP 2025 Workshops*, ser. Lecture Notes in Computer Science, vol. 15955, M. Törngren, B. Gallina, E. Schoitsch, E. Troubitsyna, and F. Bitsch, Eds., Springer, Cham, 2026, pp. 267–280. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-032-02018-5_20

[4] J. Hasan, "Optimizing Large Language Models through Quantization: A Comparative Analysis of PTQ and QAT Techniques," 2024. [Online]. Available: https://arxiv.org/html/2411.06084v1

[5] S. Ye *et al.*, "FLASK: Fine-Grained Language Model Evaluation Based on Alignment Skill Sets," 2024. [Online]. Available: https://arxiv.org/pdf/2307.10928

[6] L. Zheng, W. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, I. Stoica, " Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena " 2023. [Online]. Available: https://arxiv.org/pdf/2306.05685

[7] T. Nguyen, T. Nguyen, " An Evaluation of LLMs Inference on Popular Single-board Computers " 2025. [Online]. Available: https://arxiv.org/pdf/2511.07425